Time domain	Laplace domain	ROC	Poles and zeros
$x^*(t)$	$X^*(s^*)$	No change	Conjugated
$x(-t)$	$X(-s)$	Flips	sign changes
$ax_1(t)+bx_2(t)$	$aX_1(s)+bX_2(s)$	$R_1 \cap R_2$	$\{P_1\} \cup \{P_2\} \rightarrow$ poles
$x(t-\theta)$	$X(s)e^{-j s\theta}$	No change	Not rational
$x(t)\,e^{\theta t}$	$X(s-\theta)$	$R+Re\{\theta\}$	$\{P\}+\theta$, $\{Z\}+\theta$
$x(ct)$	$X(s/c)/\lvert c\rvert$	ROC scaled	$c\{P\}$, $c\{Z\}$
$x(t) * y(t)$	$X(s)\,Y(s)$	$R_1 \cap R_2$	$\{P_1\} \cup \{P_2\}$ $\{Z_1\} \cup \{Z_2\}$
$d^n x(t)/dt^n$	$s^n X(s)$	larger	zero at origin
$\int_{-\infty}^{t} x(\tau)\,d\tau$	$X(s)/s$		pole at origin

→ ILT $\dfrac{d^n X(s)}{ds^n} = (-t)^n x(t)$

→ ILT $\dfrac{1}{(s+a)^n} = \dfrac{t^{n-1}}{(n-1)!} e^{-at} u(t)$

→ Transfer function: $H(s) = Y(s)/X(s)$

→ causal: ROC is right sided

→ Stable: ROC includes $s=0$

→ A causal system with rational transfer function is stable iff all poles are on left half of s-plane

→ initial value th: $x(t)=0$ for $t<0$

$$\lim_{s\to\infty} sX(s) = x(0)$$

$$X(s) = \int_0^\infty x(t)e^{-st}dt \Rightarrow \lim_{s\to\infty} s\int_0^\infty x(t)e^{-st}dt$$

$$\Rightarrow \lim_{s\to\infty} s\int_0^\epsilon x(t)e^{-st}dt + s\int_\epsilon^\infty x(t)e^{-st}dt = \lim_{s\to\infty} s\int_0^\epsilon f(t)e^{-st}dt$$

goes to 0 as $t\to\infty$

$$= \lim_{s\to\infty} f(0^+)\, s\int_0^\epsilon e^{-st}dt = f(0^+)$$

→ final value th: $x(t) = 0$ for $t<0$

$$\lim_{s\to 0} sX(s) = x(\infty)$$